

Question:

Kylar went to the store to buy glasses for his new apartment. One glass costs \$5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?

Reasoning + Answer:

<think> Okay , so Kylar wants to buy 16 glasses , with the first glass costing \$ 5 , and subsequent glasses costing 60 % of the original price . That means the second glass will cost $5 * 0.6 = \$3$, the third glass will cost $3 * 0.6 = \$1.80$, and so on . To find the total cost , I need to calculate the cost of each glass and then add them up . The first glass costs \$ 5 , the second costs \$ 3 , the third costs \$ 1.80 , the fourth costs \$ 1.20 , and so on . This is a geometric sequence , where each term is 60 % of the previous term . To find the total cost , I'll use the formula for the sum of a finite geometric series : $S = a * (1 - r^n) / (1 - r)$, where S is the sum , a is the first term , r is the common ratio , and n is the number of terms . In this case , $a = 5$, $r = 0.6$, and $n = 16$. Plugging these values into the formula , I get $S = 5 * (1 - 0.6^{16}) / (1 - 0.6)$. Now I need to calculate the value of this expression .</think><answer>3.67.

